DAY 02 • STUDENT ACTIVITIES

Computing and Network Foundations for Threat Monitoring

Operating systems, network architecture, web servers and scripting — the knowledge base behind every alert

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats (with Common Competency ICT311203 Perform Computer Operations)
Coverage	Element 1 Required Knowledge 1.1–1.4: Windows, macOS and Linux; web server and website architecture; scripting languages; LAN, WAN and port forwarding
Duration	8 hours (1 training day)

Trainee Name	Batch / Section	Date

Activities Overview

No.	Activity	Type	Time	Assets Needed
2.1	Windows and Linux Command Lab	Individual (hands-on)	90 min	CTM Virtual Lab: your WKS-LAB and ubuntu-lab hosts
2.2	Reading a Network Topology	Pair	45 min	lab_network_topology.txt
2.3	Web Access Log Investigation	Individual	60 min	web_access.log
2.4	Read and Run a Monitoring Script	Individual (hands-on)	45 min	log_counter.py, web_access.log

CAUTION — Lab Rules

- Work only on the lab machines, virtual machines and files provided by your trainer.
- Never scan, probe or test any system outside the authorized training network.
- Treat all sample logs and data as confidential, as if they belonged to a real client.

Activity 2.1 — Windows and Linux Command Lab

Type	Individual (hands-on)
Time Allotment	90 min
Objective	Use OS commands to collect system, process and network information from a lab host.
Assets / Materials	CTM Virtual Lab: your WKS-LAB workstation (PowerShell tab) and ubuntu-lab VM (Ubuntu tab)

Instructions

1. Open the CTM Virtual Lab, sign in with your name and seat number, and choose Activity 2.1. Use the PowerShell tab (your WKS-LAB workstation).
2. Run each command in the table and record the requested information.
3. Repeat the equivalent commands in the Ubuntu tab (your ubuntu-lab VM).
4. Answer the analysis questions.

Work Space

Information	Command Used (Win)	Result (Win)	Command Used (Linux)	Result (Linux)
IP address				
Default gateway				
Number of listening ports				
3 running processes				
OS version				
Logged-in user				

Analysis: Which listening ports did you find on each host? Which service do they belong to? Is any of them unexpected?

Deliverable

Completed command table and analysis (Activity 2.1 worksheet in your Lab Notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Used the correct commands on both operating systems	☐	☐
Recorded accurate results	☐	☐
Identified the service behind each listening port	☐	☐

Activity 2.2 — Reading a Network Topology

Type	Pair
Time Allotment	45 min
Objective	Identify network zones, critical assets and risky port forwarding rules.
Assets / Materials	assets/lab_network_topology.txt

Instructions

1. Open assets/lab_network_topology.txt.
2. Identify the LAN, DMZ and WAN segments and the devices in each.
3. Review the firewall port forwarding rules at the bottom of the file.
4. Mark each rule as Acceptable or Risky and explain.

Work Space

Zone	Network Range	Devices

Rule No.	Acceptable / Risky	Reason

Deliverable

Zone table and rule assessment (Activity 2.2 worksheet in your Lab Notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Correctly identified all three zones	☐	☐

Criteria	Yes	No
Identified the RDP and database forwarding rules as risky	☐	☐
Explanations reference exposure to the internet	☐	☐

Activity 2.3 — Web Access Log Investigation

Type	Individual
Time Allotment	60 min
Objective	Read web server log entries and spot suspicious requests.
Assets / Materials	assets/web_access.log

Instructions

1. Open assets/web_access.log in a text editor (or `cat`/`less` on Linux).
2. Identify the normal-looking requests and the suspicious ones.
3. For each suspicious entry, record the IP, what was attempted, the status code and whether it succeeded.

Work Space

Line	Source IP	What Was Attempted	Status	Succeeded?

Deliverable

Suspicious request table (Activity 2.3 worksheet in your Lab Notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Found the SQL injection, path traversal, scanner and web shell entries	☐	☐
Correctly interpreted status codes	☐	☐
Flagged the successful web shell request (200) as most urgent	☐	☐

Activity 2.4 — Read and Run a Monitoring Script

Type	Individual (hands-on)
Time Allotment	45 min
Objective	Explain what a script does before running it, then run it safely.
Assets / Materials	assets/log_counter.py; assets/web_access.log

Instructions

1. Open assets/log_counter.py and read it WITHOUT running it.
2. Answer: What does it read? What does it change? Where does it connect?
3. In the Ubuntu tab of the CTM Virtual Lab, run: `python3 log_counter.py web_access.log`
4. Compare the output with your findings in Activity 2.3.

Work Space

What does the script read?

What does it change / connect to?

Top 3 IP addresses by request count and the status codes they received:

Deliverable

Script analysis and output interpretation (Activity 2.4 worksheet). Then press Export notebook and submit the file.

Performance Criteria (Trainer Use)

Criteria	Yes	No
Correctly explained the script before running it	☐	☐
Ran the script successfully	☐	☐
Related the output to suspicious IPs	☐	☐

Self-Check

A. Multiple Choice

Encircle the letter of the best answer.

1. Which Windows Event ID indicates a failed logon?
    a) a. 4624
    b) b. 4625
    c) c. 4688
    d) d. 1116
2. Remote Desktop Protocol normally uses port:
    a) a. 22
    b) b. 445
    c) c. 3389
    d) d. 443
3. Which is a private IP address?
    a) a. 8.8.8.8
    b) b. 10.10.5.20
    c) c. 203.0.113.7
    d) d. 1.1.1.1
4. A network segment for public-facing servers is called a:
    a) a. VLAN
    b) b. DMZ
    c) c. WAN
    d) d. VPN
5. Which PowerShell option is a common sign of malicious use?
    a) a. Get-Help
    b) b. -EncodedCommand
    c) c. Get-Date
    d) d. -Verbose

B. True or False

Write TRUE if the statement is correct, FALSE if it is not.

No.	Statement	Answer
1	HTTP status 200 means the request succeeded.	
2	Port forwarding to an internal database server from the internet is a good practice.	
3	The `ss -tunap` command lists network connections on Linux.	
4	A User-Agent of "sqlmap" suggests an automated attack tool.	

C. Short Answer

1. Explain port forwarding in your own words and give one risky example.

2. List the five steps of a web request from browser to database.

Reflection

What was the most important thing you learned today, and how will you apply it in a Security Operations Center?

Trainer Sign-off

Overall Result	Trainer Remarks	Trainer Signature / Date
☐ Competent		
☐ Not Yet Competent		